Lesson 11: Wave After Wave

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: clearing a wave of enemies no longer means the game is over - it means the NEXT wave begins. Clear all 3, and that's when you really win.

Open your own [YourName]Game project from Lesson 10 - don't start a new one.

Tip: no new assets this week - multiple waves reuse your exact existing enemy spawn logic. This completes the Level 2 Complete milestone!

Setup

1. Open your [YourName]Game project from Lesson 10.
2. Create two new Variables: wave, default value 1, and total_waves, default value 3.
3. Create a new Custom Event named SpawnWave.

Level 1 - Minimal (MVP)

Clearing a wave's enemies spawns a fresh wave, up to 3 total. Clearing wave 3 triggers the real win. The current wave number is shown on screen.

4. Move the enemy-reviving/repositioning actions out of your ResetGame Custom Event and into the new SpawnWave Custom Event instead - just the revive/reposition steps, nothing else.
5. In ResetGame, add wave=1 to your resets, and replace the old revive/reposition actions with a single Call Custom Event(SpawnWave).
6. On the enemy Actor Template's "Collided with actor" trigger, revise the win-check into an if/else (code below): if this was the last wave, win as before; otherwise, advance the wave, reset enemies_remaining, and call SpawnWave again.
7. Add a wave readout to your Interface (a third Text/Graphic Field, bound to the wave Variable with Update Real-time ticked) alongside score and lives.
8. Save, then Play/Test and play through - clearing wave 1 or 2 should revive a fresh set of enemies instead of ending the game; clearing wave 3 should show "You Win!".

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Custom Event: SpawnWave ->
  Actor – Destroy/Revive Actor(enemy_1, Revive) -> Actor –
Teleport(enemy_1, [starting position])
  (repeat for enemy_2, enemy_3, enemy_4, enemy_5)

Custom Event: ResetGame (revised) ->
  Variable – Modify Value(score, Set, 0)
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  Variable – Modify Value(lives, Set, 3)
  Variable – Modify Value(wave, Set, 1)
  Variable – Modify Value(enemies_remaining, Set, 5)
  Variable – Modify Value(game_active, Set, 1)
  Call Custom Event(SpawnWave)

enemy Actor Trigger: Collided with actor (revised) ->
  If(game_active = 1)
    Variable – Modify Value(score, Add, 10)
    Variable – Modify Value(enemies_remaining, Subtract, 1)
    If(enemies_remaining <= 0)
      If(wave >= total_waves)
        Variable – Modify Value(game_active, Set, 0)
        Message Box("You Win! - Press R to Restart")
      Else
        Variable – Modify Value(wave, Add, 1)
        Variable – Modify Value(enemies_remaining, Set, 5)
        Call Custom Event(SpawnWave)
    Actor – Destroy/Revive Actor(relate, Destroy)
    Actor – Destroy/Revive Actor(this, Destroy)
```

Level 2 - Complete (Alpha/Beta)

Wave transitions are seamless and correct every time - no double-spawns, no skipped waves, exactly 3 waves before the real win, and restarting always returns to wave 1.

- Play a full 3-wave game: confirm waves 1 and 2 each revive a fresh set of enemies with no win message, and only wave 3 triggers "You Win!".
- Let lives run out partway through wave 2 and confirm "Game Over" still works correctly, not a wave transition.
- Press R after a win or loss and confirm wave resets to 1, along with score and lives.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: briefly show a "Wave 2!" Message Box (or a dedicated Interface Field, shown for a moment) distinct from the win/lose Message Box.
- Structure: notice SpawnWave is now called from two moments - ResetGame, and the win-check's Else branch. The actual revive/reposition actions only exist once.

If Something's Wrong

- Wave 1 immediately shows "You Win!": confirm the nested If checks wave against total_waves, not just enemies_remaining alone.
- New waves never revive: confirm Call Custom Event(SpawnWave) is actually inside the Else branch, and that SpawnWave's actor names match your actual enemy copy names exactly.
- Restarting leaves wave at whatever it was: confirm ResetGame explicitly sets wave to 1.

Want More?

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